

PROJECT OUTLINE & SPECIFICATION

1. **Title** – A Preliminary Investigation into Quantum Machine Learning Algorithms for Medical Diagnostics.

2. **Aim** – The project investigates the potential of quantum (QML) machine learning algorithms in medical diagnosis and data analysis, with respect to current classical methods. Its principle aim is to evaluate whether quantum-enhanced models that utilise Quantum Support Vector Machines or Variation Quantum Classifiers (QSVMs and VQCs respectively) can offer computational advantages over conventional ML methods, with the focus hence being on implementing, simulating and benchmarking QML algorithms on publicly available medical data, against current classical algorithms.

3. **Objectives** – The project contains five principle objectives that together, should investigate and demonstrate the prospects of current quantum research, hardware and software.

A. **Literature Review** – Review existing quantum and classical algorithms and their applications in healthcare, identifying their key techniques and performance qualities.

B. **Implementation of Research** – Develop a simulation framework in Python using Qiskit to implement at least one, however preferably two or more, QML algorithms on classical hardware.

C. **Medical Application** – Apply QML models to at least one, however preferably two or more, medical datasets.

D. **Comparative Evaluation** – Benchmark QML models against their classical counterparts in terms of their most significant qualities in medical diagnostics (eg speed, reliability, efficiency, generalisability).

E. **Analysis & Report** - Analyse the result to determine where QML stands with respect to current classical approaches at the moment, the potential differences in simulations and implementation on real hardware, the prospects of this academic and industrial area, and the implications this has for future QML integration in medicine.

4. **Deliverables** – The project's deliverables in essentially based on the objectives, and how they can be best realised in an academic setting.

A. **Literature Review Chapter** – Review of QML fundamentals, current algorithms and their medical applications.

B. **QML Simulation Frameworks** – Utilisation of Python libraries (Qiskit) to simulate benchmark-able QML algorithms on classical hardware.

C. **Medical Application Prototype** – QML model trained and tested on open medical datasets.

D. **Benchmark Software & Report** – Comparison of QML and classical ML models on software that displays values of diagnostically desirable qualities.

E. **Dissertation** - Written report detailing, discussing and evaluating total findings, prospects and recommendations.

5. **Plan** – Due to the investigatory nature of the project, the plan will take the form mainly as a schedule of deliverables to track the progress of the project, and to make adjustments accordingly. A schedule/plan isn't the route, nor the map of the progress, however is created to be used as a guide. The project consists of three phases:

A. **Phase I** – 4.A + 4.B · 6 weeks · 10 Nov – 12 Dec 2025

Week	Dates	Object	Activity
7	10-16 Nov 2025	3.A	Initial literature investigation, gathering reliable and quality sources.
8	17-23 Nov 2025	3.A	Investigation of reading sources, and creation of "stack" of relevant literature for future reading and reference.
9	24-30 Nov 2025	3.A	Further Reading
10	1-7 Dec 2025	3.A	Summary of reading and literature review.
11	8-14 Dec 2025	3.B	Python environment set-up, Qiskit installation, initial coding framework
12	15-21 Dec 2025	3.B	Begin implementation of first QML algorithm

B. Phase II – 4.C + 4.D · 9 weeks · 26 Jan – 27 Mar 2026

Week	Dates	Object	Activity
1	26 Jan-1 Feb 2026	3.B	Complete first QML algorithm implementation
2	2-8 Feb 2026	3.B	Implement second QML algorithm and classical ML counterparts
3	9-15 Feb 2026	3.C	Dataset acquisition, preprocessing, and exploratory analysis
4	16-22 Feb 2026	3.C	Train QML models on first medical dataset; begin second dataset if applicable
5	23 Feb-1 Mar 2026	3.D	Design benchmark metrics (speed, reliability, efficiency, generalisability)
6	2-8 Mar 2026	3.D	Run comprehensive benchmarks on QML vs classical models
7	9-15 Mar 2026	3.D	Collect final results, validate findings, document performance data

C. Phase III – 4.E · 2 weeks · 27 Apr – 08 May 2026. Last two weeks devoted to dissertation write-up, proofreading and reporting.

6. Risk – The project is ambitious, and works in relatively new areas using computationally expensive methods.

A. Prioritization strategy – Objectives are ordered by dependency and importance. 3.A and 3.B are the highest priorities as they enable all subsequent work, and hence are aimed to be completed in S1. If time constraints arise, 3.D will be simplified to focus on fewer classical baselines rather than comprehensive comparison across multiple algorithms.

B. Early prototyping – By end of S1, a QML simulation framework is aimed to be established, providing early proof-of-concept and identifying technical challenges with quantum simulation.

- C. **Dataset contingency** – As many candidate medical datasets will be identified as possible during literature review. If one dataset proves unsuitable for QML simulation, alternative publicly available datasets will be used.
 - D. **Computational limitations** – QML simulations may be computationally expensive for larger circuits. Small qubit counts, and scaling up will be done only if feasible. Cloud-based quantum simulators (IBM Quantum, Google Colab) will be investigated.
 - E. **Time buffers** – The plan includes many formal and informal buffers in for unexpected implementation challenges, simulation convergence issues, or additional experiments requested by assessor.
 - F. **Modular development** – Software will be developed incrementally with version control, allowing partial submission if complete implementation of all QML algorithms isn't achieved.
7. **Ethics** – There is no collection of data that is private/unclear of consent since this project utilises publicly available data. This project is highly experimental as well, and isn't in service of medical applications.
- A. **Data Handling & Anonymisation** - Exclusive use of publicly available data, that neither can be identified back to persons, or will be displayed in any form of output. The data used must have already undergone ethical approval for research, with verification of data use agreements pertaining to academic research.
 - B. **Reporting Transparency** – Limitations of quantum simulation on classical hardware will be clearly stated. Results, on what will be limited datasets, will emphasize that this is solely simulation based research, and not an actual quantum hardware implementations that may differ from results on real quantum hardware. QML technology is in early stages and all results will be exploratory, as well as any claims about advantages over classical methods will be present with the caveats to avoid misinterpretation in clinical contexts.
 - C. **Usage** – This project involves secondary analysis of existing datasets with no direct human participation, testing or data collection.